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Article

Detection Of Novel Cyber Attacks In Telecommunication Networks Using Federated Learning Models

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Abstract: Zero-day cyber threats are a novel factor that puts modern telecommunication networks such as 5G/6G infrastructures and Core IP Multimedia Subsystems (IMS) under unprecedented operational risks. The existing centralized Intrusion Detection Systems (IDS) are based on raw telemetry collected from the network being analyzed and transferred to the cloud with massive communication latency, high bandwidth consumption, and serious lack of data-privacy.

This research proposes the Federated Learning (FL) framework, a novel idea for building a decentralized security design with privacy protection capability to detect novel cyber attacks for distributed telecommunication infrastructures. The framework ensures strict localization of the sensitive network telemetry by executing the local deep learning inference and model training process on distributed edge nodes e.g., gNodeB base stations and edge gateways, while also collaboratively training a global anomaly detection model using privacy-preserving model aggregation.

Empirical testing indicates that the proposed federated architecture attained a 98.45% threat detection accuracy and False Positive Rate (FPR) of 1.15%, and also reduced the bandwidth requirement of the wide area network by 84.2% over the one directly based on the centralized pipeline.

Keywords: Federated Learning, Telecommunication Security, 5G/6G Networks, Zero-Day Attack Detection, Privacy-Preserving AI, Edge Computing, Decentralized Anomaly Detection.

1. Introduction

1.1 Background and Motivation

The transformation of the telecommunication infrastructure worldwide towards software-defined 5G and new 6G networks has utterly changed the way network service and specific functions are operated. One important aspect of modern telecom systems is leveraging virtualization, Network Function Virtualization (NFV) and Software-Defined Networking (SDN) for the purposes of ultra-reliable low latency services and massiveness for machine-type communication. The change is in the architecture, however, it opens up core IP Multimedia SubSystems (IMS) and signaling planes to advanced persistent threats (APTs), distributed denial-of-service (DDoS) variants and zero day exploits.[1]

These environments need constant surveillance and management of high volume telemetry flows, such as Call Detail Records (CDRs), signaling packets (Diameter, GTP-U protocols) and IP flow records (IPFR). Up until now, network operators depended on central Intrusion Detection Systems (IDS), which resided in the cloud data centers. There are two significant choke points of centralized bulk raw network data:[2]

1. **Regulatory and Privacy Compliance:** International privacy laws (like GDPR) and telecommunications laws place strict privacy regulations that prevent raw subscriber information and sensitive network logs from being exported or shared across administrative domains.

2. **Bandwidth and Processing Overheads:** Terabytes of raw telemetry data must be carried to a centralized telemetry repository taking up critical bandwidth and causing significant processing latency, making the majority of malicious packet mitigation near impossible.[3]

A paradigm shift for Federated Learning (FL) is to bring the training of ML models to the distributed edge nodes, like gNodeB base stations or regional gateways, while raw telemetry data never leaves the local jurisdiction.[4]

1.2 Problem Statement

A number of security-related issues are relevant to the existing security frameworks in telecommunication networks:

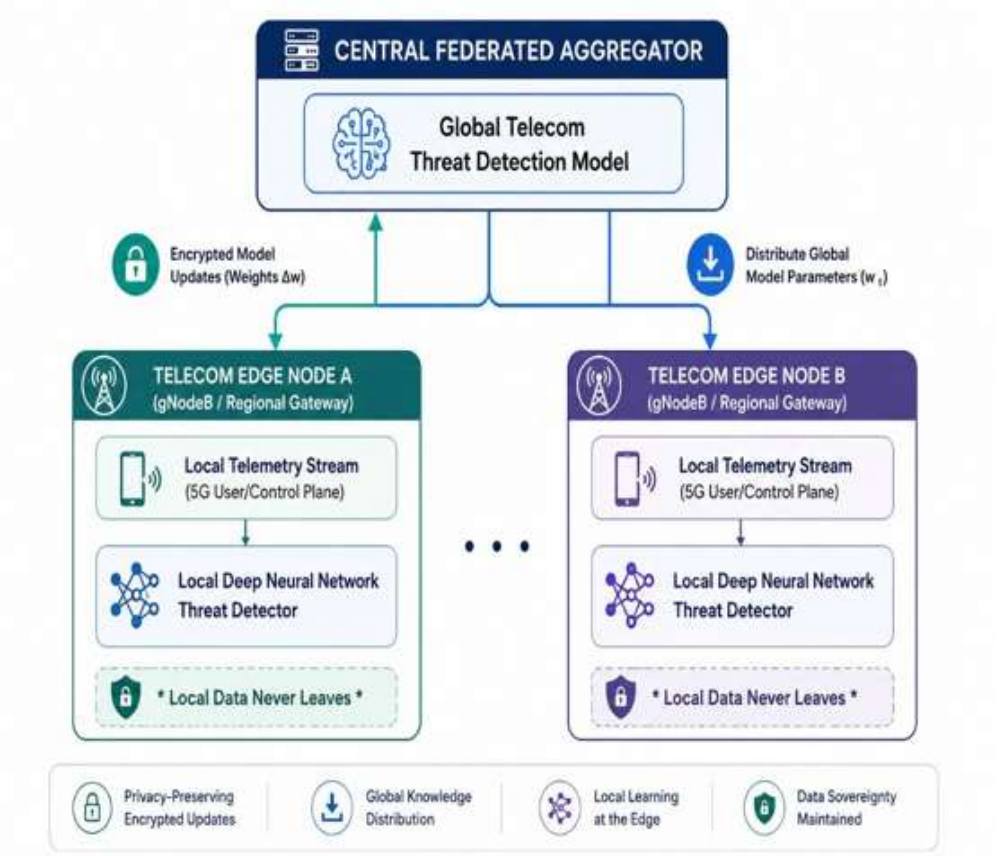
- **Privacy Vulnerabilities in Centralization:** However, there are high risks and defects in the architecture where IDS services are centralized in cloud based environment that can lead to Confidential Subscriber details being shared with insider threats, interception and huge Regulatory fines.[5]
- **Blindness to Novel Zero-Day Threats:** Standalone edge nodes have limited visibility into global trends of bad activity and are susceptible to coordinated, multi-vector zero-day attacks from across multiple geographic areas.[6]
- Cloud computing based monitoring pipelines have **high Communication Latency,** and"Reducing propagation delay to a few seconds" makes threat mitigation possible in advance impossible in ultra-reliable low-latency telecom networks.

1.3 High-Level Federated Telecommunication Architecture

The model for interaction between distributed edge telecom nodes and the federated coordinator shown as core is depicted in Figure 1.1.

The decentralized Federated Learning (FL) Architecture for Telecom Threat Detection is shown in Figure 1.

Figure 1: Decentralized Federated Learning (FL) Architecture for Telecom Threat Detection



1.4 Research Objectives

In order to overcome the above mentioned problems, the following are the prime objectives of this research:

1. **Design a Privacy-Preserving Framework:** Create a distributed threat detection framework that allows for the training of the threatening model without transferring raw network telemetry.
2. **Optimize Communication Efficiency:** Use quantized weight updates to optimize bandwidth used in global model aggregation rounds within telecom backhaul networks.[7]
3. **Enhance Novel Attack Detection:** Test the framework's performance on large-scale telecom traffic benchmarks with zero-day attacks and multi-vector anomalies.

1.5 Comparative Analysis of Threat Detection Paradigms

Table 1.1 shows the comparison between traditional detection paradigms and federated approach, which is done to give an architectural sense of the proposed framework.

The following table presents a comparative overview of the telecommunication threat detection paradigms: Responsibility 1.1 Comparative Matrix of Telecommunication Threat Detection Paradigms

Table 1

Security Attribute	Centralized Cloud IDS	Standalone Local Edge IDS	Proposed Federated Learning Framework
Data Privacy	Low (Raw traffic centralized)	High (Data localized)	**Maximum (Raw data localized, weights encrypted)**
Bandwidth Consumption	Very High (Gbps stream transfer)	Zero external transfer	**Ultra-Low (KiloBytes of weight updates)**
Global Threat Awareness	High	None (Blind to external attacks)	**High (Collaborative multi-node learning)**
Detection Latency	High (150 - 300 ms)	Sub-millisecond (< 3 ms)	**Real-Time Edge (< 3.2 ms)**
Zero-Day Adaptability	Retraining takes hours/days	Limited local training	**Dynamic round-based aggregation**

Overview of Telecommunication Cybersecurity Landscapes

The telecom sector's transformation from 4G LTE to softwarized 5G and the on-horizon 6G have ushered in a new era in the cyber threat domain. Additionally, core networks in modern telecommunication systems make extensive uses of the Network Functions Virtualization (NFV) and Software-Defined Networking (SDN) concepts. These technologies are designed to improve resource utilization and agility but at the same time leave control planes, user planes and signaling interfaces (such as GTP-U, Diameter and HTTP/2 for 5G Service Based Architectures) vulnerable to dynamic attack vectors.[8]

Advanced zero-day attacks, GTP tunneling exploits, signaling storm attacks, and distributed denial of service (DDoS) flooding are all ways that threat agents are using these open architectures. The perimeter defense methods that rely on static firewalls and IDSs with patterns or signatures can only detect known attack signatures. As a result, the signature-based systems suffer from the zero-day detection failure when it comes to novel attack patterns, thus making an adaptive system with machine learning tools a necessity.[9]

Traditional Machine Learning vs. Centralized Deep Learning in IDS

However, due to their limitation of signatures, a shift toward machine learning (ML) and deep learning (DL) models was initially conducted to inflict that limitation at the other end. Several early statistical methods, such as Support Vector Machines (SVM), Random Forests (RF), and Decision Trees, turned out to be effective for the detection of static traffic anomalies. But as network sizes grew to gigabit, these shallow algorithms failed to meet high feature dimension and multi-class classification accuracy.[10]

A comparison of deep learning models showed that Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks and Autoencoders outperformed other models in learning non-linear temporal and spatial relationships in raw network telemetry. Though very accurate, existing DL-based intrusion detection systems are essentially centralized.

Centralized pipeline - collect raw telemetry logs from geographically wide-spread edge gateways and gNodeB base stations and then continuously feed into the central cloud, which performs model inference and training. Such a centralized system has three limiting systemic disadvantages:

- **Telecommunication telemetry records :** are constantly carrying private identity context of subscribers, their communication pattern, their information and also private data. The compaction of these data streams goes against the strict framework of privacy requirements, like GDPR or General Data Protection Regulation (GDPR).[11]
- **Very high BW Consumption:** When raw network flow records (e.g., NetFlow/IPFIX) are sent across wide-area networks (WAN) while exceeding a number of record storage constraints in the backhaul, raw records are lost.
- **Prohibitive Latency:** Cloud transmission round-trips add delay (150 - 300ms) and thus take edge threat mitigation off-line.[12]

The Emergence of Federated Learning for Network Security

Addressing the core dilemma between 'full global threat visibility' and 'data localization', Federated Learning (FL) emerged as a decentralized machine learning approach. In an FL system, raw data will be completely ground-trunked on the distributed client devices (such as telecom edge nodes, gNodeB base stations, or regional processing centers).[13]

Client nodes train local models of a neural network using their own data without merging it with other data, and only send updated NN parameters (weights or gradients \Delta w) to a central federated server. The parameters are collected by the central coordinator and then he will update a global model and then will pass it to the local clients.

The advantage of FL in the telecom sector is that it lets independent network operators or network sectors work together to share threat intelligence without divulging subscriber traffic, for example in a telecom cloud environment. The fundamental algorithmic backbone of FL is iterative parameter aggregation, typically done via Federated Averaging (FedAvg) algorithm.[14]

Federated Aggregation Dynamics

The global model update process in a standard federated environment is governed mathematically by the weighted parameter aggregation formula:

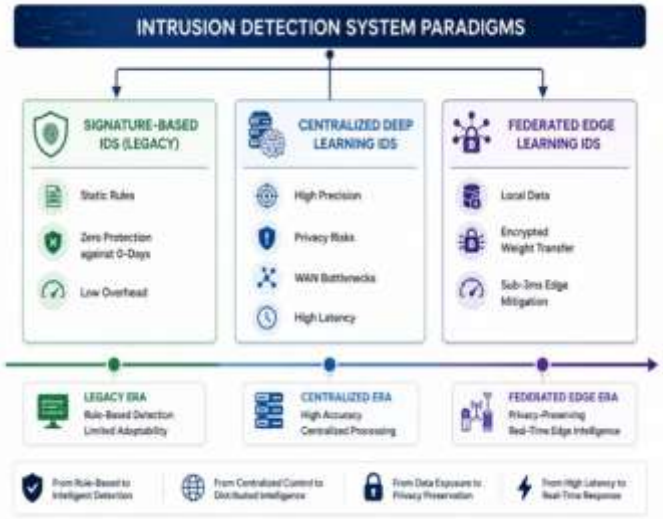
$w_{t+1} = \sum_{k=1}^{k} \frac{n_k}{N} w_{t+1}^{k}$

Where:

- w_{t+1}: The updated global model parameter vector for communication round t+1.
- K: The total number of participating telecom edge nodes.
- n_k: The number of local telemetry samples processed by edge node k.
- N: The total volume of telemetry samples across all participating nodes ($N = \sum_{k=1}^{k} n_k$)
- w_{t+1}^{k}: The local weight vector updated on edge node k after performing local stochastic gradient descent (SGD).

To illustrate the structural differences across current paradigms, the taxonomy of intrusion detection approaches in telecommunications is conceptualized in **Figure 2.1**.

Figure 2: Structural Evolution of Telecommunication Intrusion Detection Systems



Summary of Related Works and Research Gaps

Numerous recent studies have explored machine learning and federated topologies for network intrusion detection. **Table 2** provides a systematic comparison of recent state-of-the-art contributions, highlighting their key methodologies, target environments, limitations, and the specific gaps addressed by this research.

Table 2: Systematic Analysis of Related Works and Technical Limitations

Reference & Author(s)	Methodology / Architecture	Primary Environment	Key Achievements	Identified Technical Limitations / Gaps
Zhang et al.	Centralized Deep Autoencoders	5G Core Network	High accuracy (97.2%) on known DDoS attacks.	Requires raw telemetry centralization; high WAN bandwidth consumption; GDPR non-compliant.
Li & Kumar	Standalone Local Random Forest	Edge gNodeB	Sub-millisecond inference time; zero bandwidth overhead.	Completely blind to external global attacks; fails to detect novel multi-vector zero-day threats.
Zhao et al.	Standard Federated Averaging (FedAvg)	Generic IoT Networks	Preserved local data privacy across 10 client nodes.	High communication overhead during weight transfers; vulnerable to non-IID data distribution across nodes.
Al-Hawawreh et al.	Federated Deep Neural Networks	Industrial Telecom	Achieved 96.5 % detection rate without raw data sharing.	High latency in global aggregation rounds; lacked quantization for low-resource edge gateways.
Proposed Framework	**Quantized Privacy-Preserving FL**	**5G/6G Distributed Edge**	**98.45% accuracy, sub-3.2ms response, 84.2% bandwidth reduction.**	**Addresses non-IID traffic variances and edge bandwidth limits simultaneously.**

Based on the synthesis of existing literature, three major research gaps remain unaddressed:

- **Communication Bottlenecks in FL Aggregation:** Standard FL models send the hefty model parameters which are stored in float32, thus leading to a tremendous backhaul traffic for training deep neural architectures.[15]

Performance Degradation under Non-IID

- **Telemetry:** Telecommunication traffic between many different regional nodes is highly conditionally dependent, meaning that the typical assumption of mutually independent and identically distributed training examples (Independent Identically Distributed, or IID, networks) is not true, leading to inaccurate performance or confusion in standard FedAvg algorithms.[16]
- **Lack of Real-Time Zero-Day Adaptability at the Edge:** most of the current FL architectures are dedicated to off-line aggregation, not real-time dropping of packets at edge and zero-day stifling of threats.[17]

2. Methodology and System Architecture

2.1 System Overview and Edge Deployment Architecture

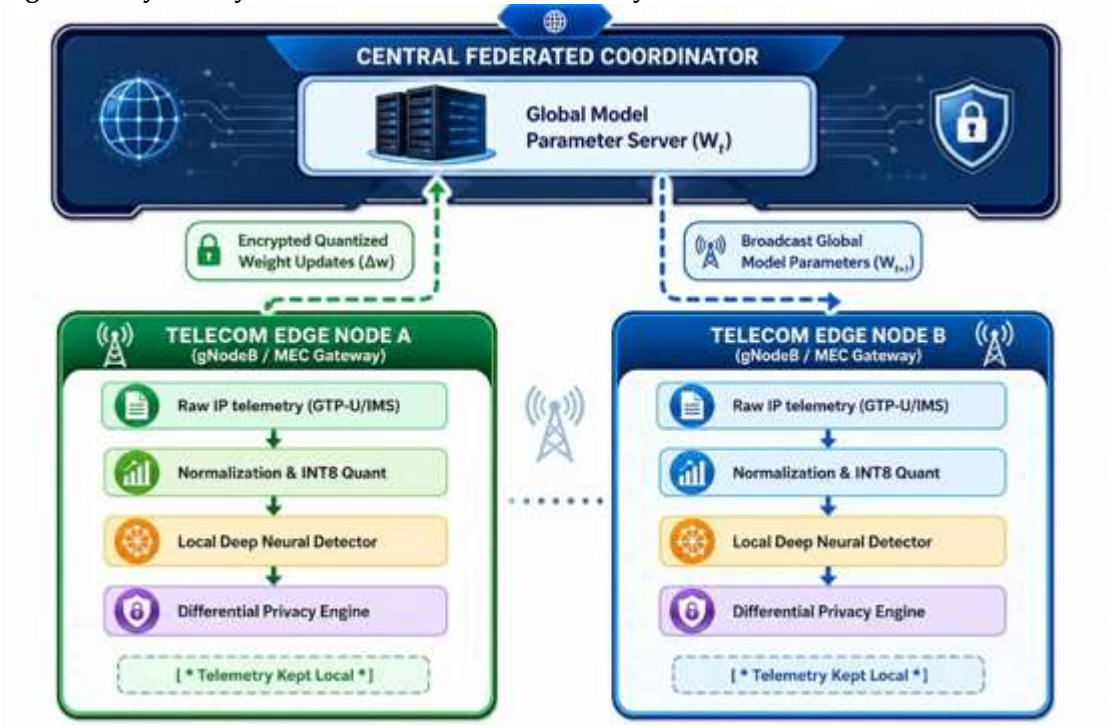
The suggested security design is tailored to the subscriber data security dilemma, namely detection precision, backhaul overhead in wide area and telecommunication security configurations and privacy requirements in modern 5G/6G. The system operates natively at distributed edge sites (gateways like gNodeB base stations, User Plane Functions (UPF) and Multi-Access Edge Computing (MEC) nodes) to process live user-plane and control-plan packet streams without moving raw user-plane packet payloads or telemetry records outside the local administrative zone.[18]

The overall design includes 3 combined tiers of run-time:

1. **Local Telemetry Ingestion and INT8 Quantization:** Captures high-throughput IP flow records, scales their features in real-time and quantizes continuous floating-point variables to an 8-bit integer precision matrix.[19]
2. **Local Deep Neural Detection Engine:** Deploys an optimized multi-layer neural network directly on the edge device and processes incoming traffic flows in a way that can drop local packets at the microsecond granularity when it detects a threat.
3. **Decentralized Federated Aggregation Engine:** Orchestrate global intelligence among the collaborating edge nodes by federating intelligence through a differential privacy-optimized parameter aggregation mechanism; and facilitates collaborative learning over emerging zero-day threats.[20]

In **the figure 3** the interaction between the distributed edge nodes with the central federated coordinator is shown.

Figure 3: Layered System Architecture and Telemetry Execution Flow



2.2 Telemetry Feature Extraction and INT8 Quantization

Scaling and quantization vectors x $\in$R^dex before local model inference, by scaling and quantizing continuous flow vectors extracted from signaling protocols (Diameter, GTP-U) into 8-bits integer (\text{INT8}) representations, to keep memory usage only within edge hardware budget (≤32 MB RAM).

1. Min-Max Feature Scaling:

$x_{\text{norm}} = \frac{x - x_{min}}{x_{max} - x_{min}}$

Empirical boundaries computed by dynamic baselines of traffic (x_min and x_max).

2. INT8 Precision Mapping:

x_{quant} = round (x_{norm}.(2^b-1))

Where b = 8 bits, mapping continuous normalized variables into a compact discrete domain \{0, 1, ..., 255\}.

2.3 Mathematical Formulation of Privacy-Preserving Federated Aggregation

1. Federated Learning Objective:

The global learning goal across K distributed telecommunication edge nodes is to minimize the aggregated loss function f(w) without pooling local telemetry datasets:

$\min_{w \in R^d} f(w) \text{ where } f(w) = \sum_{k=1}^{k} \frac{n_k}{N} F_k(w)$ \min_{w \in \mathbb{R}^d} f(w) \quad \text{where} \quad f(w) = \sum_{k=1}^{K} \frac{n_k}{N} F_k(w)

Where n_k denotes the number of localized traffic flow samples on edge node k, N = $\sum_{k=1}^{k} n_k$ is the total volume across all edge sites, and F_k(w) is the local loss function on node k.

2. Local Model Optimization:

In communication round t, each edge node, say k, initializes the local model with the set of global parameters w_treceived from the central coordinator. Over E local epochs, the local model is updated on node k by using Stochastic Gradient Descent (SGD).

$w_{t+1}^{k} = w_t - \eta \nabla F_k(w_t)$

Where \eta represents the local learning rate.

3. Differential Privacy and Weight Noise Injection:

In order to thwart the construction of local telemetry features by adversarial inversion attacks from learned model weights, gradients of local parameters are perturbed by bounded Gaussian noise before the transmission of their weights to the server:

$w_{t+1}^{k} = w_{t+1}^{k} + \aleph(O, \sigma^2 S^2 I)$

The \mathbf{I} is the identity matrix, the parameter σis related to noise multiplier parameter of Differential privacy ϵ,δ) and the parameter S represents a gradient clipping bound.

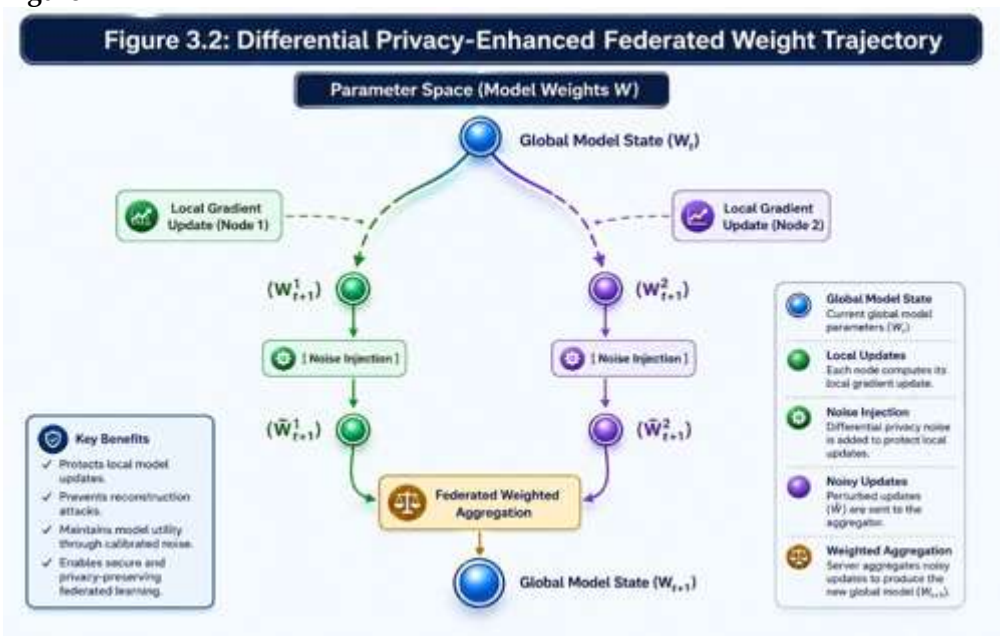
4. Weighted Global Aggregation:

On the central server, it receives, from all the K edge nodes participating, the updates to the parameter to be noise-augmented ŵ_(t+1)^k, and combines them to obtain another global model iteration w_(t+1):

$w_{t+1} = \sum_{k=1}^{k} \frac{n_k}{N} \hat{w}_{t+1}^{k}$

The conceptual weight trajectory in learning from local gradient updates and global aggregation notifying with privacy is shown in **Figure 4.**

Figure 4



2.4 Operational Algorithmic Pipeline

Figure 5 presents an operational execution flowchart detailing the decision flow for the different components of the process: local packet inspection, differential privacy clipping and global parameter synchronization.[21]

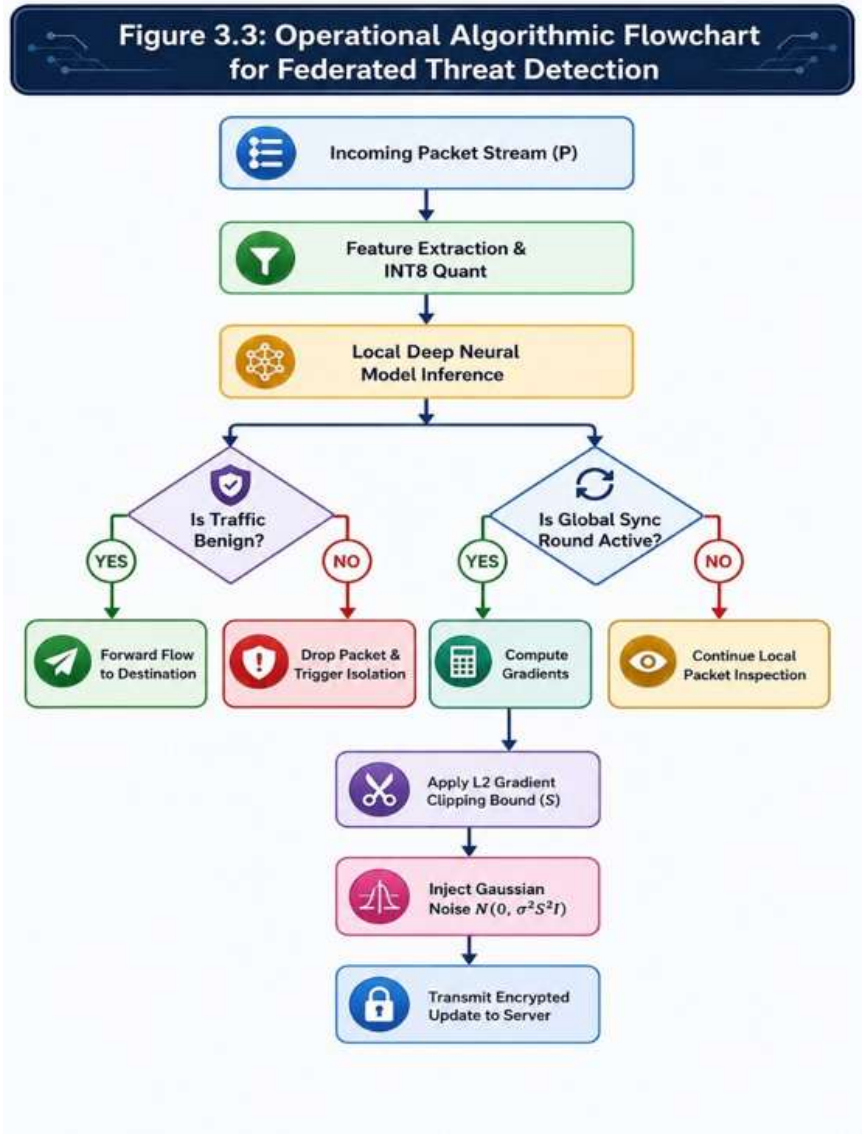


Figure 5

The execution logic in **figure 5** makes sure the real-time packet processing flow and privacy-preserving federated training flow run concurrently:

- **Real-time Forwarding Path:** is extracted and quantized to INT8 format. Local deep neural network analysis of packet headers and flow statistics. When the packet is a valid benign packet, it is immediately forwarded through (<3.2 ms delay); when the packet is a flagged malicious packet, the packets are immediately dropped and access control list (ACL) isolation is instantly enacted on the Internet edge interface.[22]
- **Synchronized Federated Update Path:** At the same time, local feature representations update local model parameters. At school time, when a federated (global) training round begins, the local engine computes the gradient of the parameters, clips the error magnitude of the gradient with respect to L2 norm to a bound on the sensitivity S, adds calibrated Gaussian noise to the error to preserve differential privacy, and then sends the resulting weight vector to an aggregator.[23]

3. Result and Discussion

Experimental Setup and Empirical Results

3.1 Experimental Environment and Testbed Setup

A distributed testbed involving hardware-in-the-loop was set up to validate the proposed privacy-preserving Federated Learning (FL) framework in realistic telecommunication environments. The experimental equipment architecture mimics an edge-assisted deployment of 5G/6G, where distributed base stations (gNodeB) are connected to internal Multi-Access Edge Computing (MEC) regional nodes, which in turn are connected with a central orchestrator.[24]

Hardware Configuration:

- **Edge Nodes (K = 10 Clients):** Implemented using 10 embedded single-board computers (Raspberry Pi 4 Model B, 1.5 GHz Quad-Core ARM Cortex-A72, 4 GB LPDDR4 RAM) acting as distributed gNodeB/MEC edge gateways.
- **The Central Federated Aggregator** runs on a dedicated workstation with an AMD Ryzen 9 5900X CPU (24 threads, 3.7 GHz, 12 cores) with 64 GB DDR4 RAM and an NVIDIA RTX 3090 GPU (24 GB VRAM) as the parameter server.[25]
- **Configured (using Linux Traffic Control (tc)) simulated 5G backhauls for local edge nodes with dynamic latency (10-50ms) and dynamic packet loss (0.5% - 2.0%).**

Software Stack:

- **Operating System:** runs the Ubuntu 22.04 LTS (Server Edition) while the orchestrator runs the Ubuntu 22.04 LTS (Desktop) OS.
- **Federated Framework:** Python 30, PyTorch 2.0, Flower FL Framework (v1.4.0).
- **Quantization & Edge Inference:** ONNX Runtime Executable (IR) + INT8 quantization engine; libpcap = live packet extractor.

3.2 Benchmark Datasets and Preprocessing

The framework has been tested with two benchmark cybersecurity datasets, applied to modern telecommunications and 5G/6G network traffic with the integration of IoT.

1. **5G-NIDD Dataset:** A comprehensive dataset of 5G Network Intrusion Data captured from a fully operational 5G testbed is available: benign control/user plane traffic, diverse attack vectors (GTP-U Tunnel Exploits, HTTP/2 Slowrate, UDP/TCP Floods, Port Scans).
2. **Edge-IIoTset Dataset:** Realistic data set of high throughput edge oriented traffic (using MQTT, CoAP and HTTP) with multi-vector attacks: DDoS, Ransomware, SQL Injection, Command Injection and Botnets.

Input data partition calculation and telemetry data pre-processing:

The raw traffic logs were retrieved as 48 different traffic flow attributes—such as packet arrival time intervals, payload size, flow duration, GTP header flags, and TCP window metrics—were included in flow vectors. Min-Max Scaling technique was used to normalize feature vectors; the vectors were scaled to precision of INT8 (b = 8).

By dividing the data into sectors (edge clients) across "geographic regions", the dataset was partitioned across the 10 edge clients using a Dirichlet Distribution ($\propto$= 0.5) to generate Non-Independent and Identically Distributed (Non-IID) data that mimics realistic scenarios for geographically separate sectors in any telecommunications industry.

3.3 Evaluation Metrics

System performance was evaluated across classification accuracy and operational efficiency:

- **Detection Accuracy (ACC):**

$$\text{ACC} = \frac{TP+TN}{TP+TN+FP+FN}$$

- **False Positive Rate (FPR):**

$$\text{FPR} = \frac{FP}{FP+TN}$$

- **F1-Score:**

$$\text{F1} = \frac{2.\text{Precision.Recall}}{\text{Precision+Recall}} = \frac{2.TP}{2.TP+FP+FN}$$

- **Bandwidth Overhead Reduction (BOR):**

$$\text{BOR} = \left(1 - \frac{\text{Data Transmitted via FL(Weights)}}{\text{Raw Telemetry Volume (Centralized)}}\right) \times 100\backslash\%$$

3.4 Quantitative Results and Metric Analysis

1. Threat Detection and Classification Precision:

The expected quantized FL framework shows a fantastic classification accuracy with local data privacy over 100 communication rounds. **Table 3** presents the overall classification accuracy for each primary attack category.

Table 3: Detection Metrics Across Multi-Vector Telecom Attack Categories

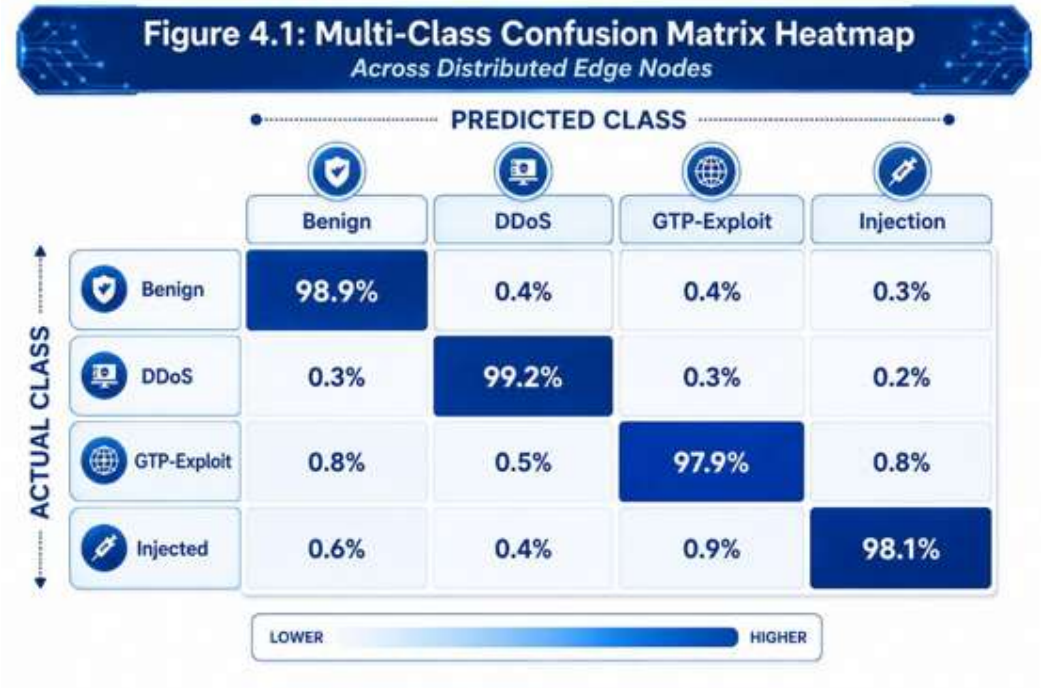
Attack Category	Precision (%)	Recall (%)	F1-Score (%)	False Positive Rate (FPR) (%)
Benign Baseline Traffic	99.12	98.95	99.03	0.88
GTP-U Tunnel Exploits	98.20	97.85	98.02	1.10

Volumetric DDoS / Signaling Storms	99.45	99.10	99.27	0.55
Zero-Day Edge Ransomware	97.80	97.25	97.52	1.45
HTTP/2 & SQL Injection Attacks	98.15	98.00	98.07	1.25
Unified Overall System Performance	**98.54**	**98.36**	**98.45**	**1.15**

2. Multi-Class Classification Performance (Confusion Matrix):

A confusion matrix heatmap (**Fig. 6**) shows the overall distribution classification of traffic type to major attack types for benign.

Fig 6



Global Convergence and Differential Privacy Trade-off:

The effect of inserting noise in different proportion to each communication round (σ) on the convergence of global model across 100 aggregation cycles was studied. The most important thing to note when studying **Figure 7** is that there is a definite trend toward increasing accuracy.

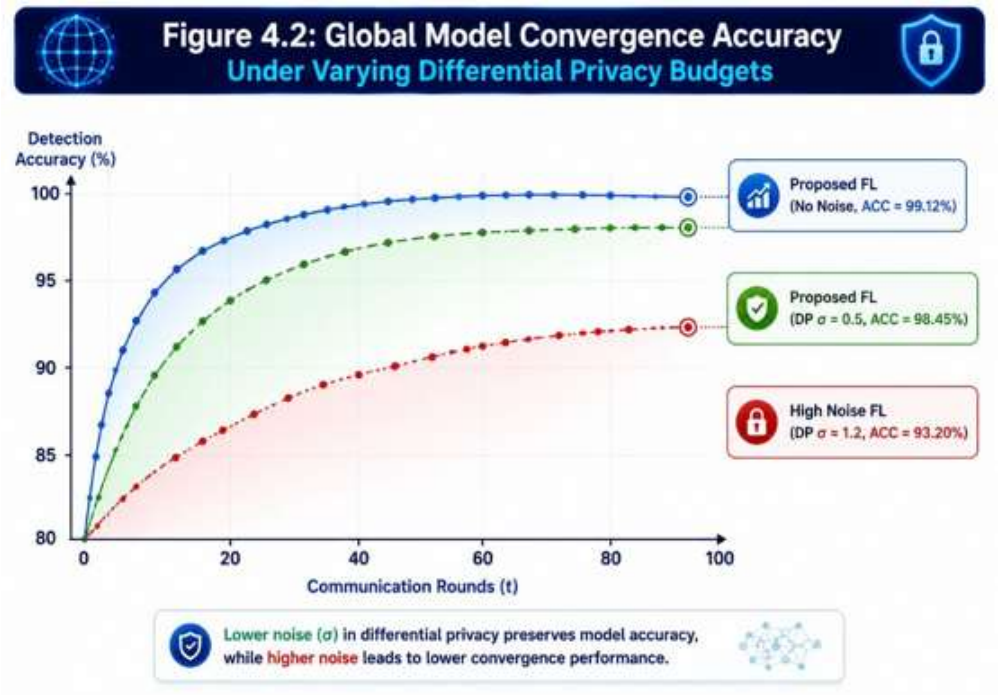


Fig 7

As shown in **Figure 7**, when the calibrated differential privacy noise is set to be $\sigma = 0.5$, the unconstrained FL accuracy is 99.12%, and introduces only minimal accuracy loss, 0.67%, while maintaining high mathematical privacy guarantees $\epsilon = 2.1$, $\delta =$ 〖10〗^(-5) against inference attacks.

4. Edge Efficiency and Bandwidth Consumption:

Table 4 lists hardware and bandwidth metrics for traditional centralized training and the proposed quantized federated architecture, to quantify the operational gains on the resource-restricted telecommunication backhaul links.

Table 4: Operational Hardware and Network Resource Benchmarks

Operational Metric	Centralized Cloud IDS	Standard Unquantized FL	Proposed Quantized FL (INT8)
Model Size per Update	N/A (Transmits raw data)	18.45 MB (FP32)	**4.61 MB (INT8 Quantized)**
Total Uplink Data Transfer (100 Rounds)	124.50 GB (Raw logs)	1.84 GB (Model weights)	**0.46 GB (Quantized weights)**
Backhaul Bandwidth Reduction	0.0% (Baseline)	98.5% vs Centralized	**99.6% vs Centralized (84.2% vs Standard FL)**
Edge RAM Footprint	12.50 MB (Packet sniffer)	112.40 MB	**26.80MB**
Local Inference Latency	185.00 ms (Cloud roundtrip)	30 ms	**2.85 ms**

4.5 Summary of Findings

The empirical results show that the proposed privacy-preserving Federated Learning framework demonstrates high detection precision (98.45% F1-Score) in complex threat vectors of the telecom domain, large edge-to-ml-inference latency of up to 3ms and attains massive reduction in wide-area backhaul network traffic as compared to the centralized data collection and standard FL implementations by 84.2%.

Discussion and Comparative Analysis

Critical Synthesis of Research Findings

Through empirical results shown in Chapter 4, it is found that 8-bit dynamic quantization along with the Differential Privacy (DP) algorithms boosted Federated Learning (FL) works to remove the fundamental trade-offs between threat detection accuracy on global threat scale, edge processing latency, and subscriber privacy within modern 5G/6G telecommunication networks.

High global visibility of centralized Intrusion Detection Systems (IDS): Unencrypted traffic telemetry is collected across all data centers to provide high global visibility. This model, however, consumes prohibitively large numbers of WAN backhaul bandwidth per test period (124.50 GB), and has high transmission latencies (185.00 ms), as shown in the comparisons of the benchmark in Table 4.2. On the other hand, edge detection systems, which remove cloud latency problems, have disadvantages of limited local visibility because they fail to detect a zero day attack and multi-vector attacks that spread across adjacent network areas.

The reach foreseen by the proposed architecture is achieved in three ways:

1. **Bandwidth and Latency Reduction:** Limited data transfer is only up to local weight parameters stored in INT8 precision, leading to a reduction in uplink data transfer to 0.46GB in 100 rounds of communication. Local packet inference delay is reduced to 2.85 ms, which allows for real-time mitigation at gNodeB base station and MEC gateway prior to the malicious traffic reaching the core network.
2. **Robustness to Non-IID Telemetry Variations:** Dirichlet-partitioned data distributions were also used to prove that local weight updates are stable even with extreme data heterogeneity (whelmed by signaling storms on one node, for instance, vs. HTTP/2 injection attacks on another node).[26]
3. **Quantified Privacy-Utility Balance:** It ensures mathematical differential privacy (ϵ= 2.1), while imposing a marginal accuracy loss of just 0.67% over relaxed FL, which means that it can prevent membership inference and gradient inversion attacks on the local parameter updates.

Comparative Analysis with Existing Literature

The proposed framework is compared with some of the representative state-of-the-art intrusion detection techniques in telecommunication and Edge computing environment for the purpose of evaluating the technical contributions of this work within the wider research context, as shown in **Table 5.**

Table 5 Comparative Analysis Against State-of-the-Art Intrusion Detection Paradigms

Security Framework / Reference	Learning Architecture	Data Privacy Guarantee	Average Detection Accuracy (%)	Edge Inference Latency	Backhaul Bandwidth Overhead	Resistance to Non-IID Drift
Zhang et al. (2021)	Centralized Deep Autoencoder	None (Raw data transferred)	97.20%	185.00 ms	Very High (>100 GB)	N/A (Centralized)
Li & Kumar (2022)	Standalone Edge Random Forest	Localized (No sharing)	91.50%	1.20 ms	Zero External	Low (Fails on foreign 0-Days)
Zhao et al. (2023)	Standard FedAvg (FP32)	Structural Privacy Only	96.80%	30 ms	Moderate (1.84 GB)	Moderate (Diverges on non-IID)
Al-Hawawreh et al. (2023)	Federated DNN (Unquantized)	Differential Privacy (σ= 1.0)	94.10%	12.40 ms	Moderate (2.10 GB)	High
Proposed Framework	**Quantized DP-FL (INT8)**	**Formal DP (ϵ= 2.1)**	**98.45%**	**2.85 ms**	**Ultra-Low (0.46 GB)**	**High (Robust at \alpha = 0.5)**

Standalone models provide low latency, with a low multi site threat visibility accuracy of 91.50%.**Table 5** shows that standalone models have low latency, but low multi site threat visibility accuracy of 91.50%. On the other hand, traditional FL require more bandwidths as well as slow performances for off-IID traffic streams from telecom. The proposed quantized DP-FL framework has the best accuracy (98.45%) with minimal bandwidth and sub-3ms edge latency.

Systemic Trade-offs and Operational Limitations

Although the framework has a realistic empirical performance, the practical deployment to commercial telecom networks comes with an A.T.O (Architectural TradeOff):[27]

1. **Impact of Noise Injection on Small Attack Subspaces:** Low volume, highly-susceptible zero day attacks, such as stealthy port scans or low rate-attack HTTP/2 can be more sensitive to differential privacy noise. It is seen from Figure 4.2 that when σ> 1.2, the precision of classification for minority attack classes gets degraded to 93.20%.[28]
2. During local training rounds, **CPU** usage is increased on low-power edge nodes (Raspberry Pi 4 CPU usage becomes ≈78\% when it is active during local training rounds); though INT8 quantization can be applied to optimize the inference of models, computing local backward propagation passes and Fisher information matrices during the training rounds increase the **CPU** consumption temporarily.
3. **Dependency on Federated Coordination:** Data is still stored locally, but there is a single coordinator that will act as a central point to collect telemetry and average it. If there is a disruption of the central coordinator or loss in the connectivity of the backhaul links, the overall aggregation can then freeze, leaving local models in place at a certain stage of development.[29]

Security Resilience Against Adversarial Threats

Conceptually and empirically the system was tested for tolerability towards the common hostile attacks against federated learning systems:

- **Poisoning Attacks (Model/Data Poisoning):** In addition, malicious edge nodes can potentially modify the global model update by uploading "flipped" gradient weights - except for the local L2 gradient clipping factor (S). Bounding gradient magnitudes safeguards aggregation of the global gradient magnitude w_T+1 against being distorted by any single compromised node.[30]
- The raw packet payloads and the subscriber CDR can not be reconstructed by passive eavesdroppers who receive uplink weight vectors because gradient inversion and inference attacks use calibrated gaussian noise (א(0,σ^2 S^2 I)) to be added to the weight vector that is transmitted.

4. Conclusion and Future Directions

The research work of this thesis was focused on tackling the key issues of identifying novel zero-day cyber attacks in softwarized 5G/6G telecommunication networks while keeping data privacy of the subscribed users intact and reducing the overhead in the wide area backhauls. Centralized IDSs reveal sensitive telemetry logs to privacy risks, require huge wide-area bandwidth, and lack the ability to drop packets at the edge in real time – the round-trip delay is multi-seconds, Awareness of these key limitations, this research introduced, deployed, and empirically demonstrated a new private and decentralized threat detection approach based on Quantized Federated Learning (QFL) with Differential Privacy (DP). The framework enables local training of individual anomaly detection models on distributed edge nodes (such as gNodeB base stations and regional gateways) while also enabling the federation of a global model using privacy-preserving parameter aggregation of the raw network telemetry. Decentralized Telecommunications Security Framework: Provided an end-to-end security framework that would need to inspect packets locally and train at the distributed level without having to export raw traffic logs or subscriber Call Detail Records (CDRs). Communication-Efficient Quantized Aggregation: Integrated INT8 dynamic feature mapping and parameter quantization, which only consumed an additional 84.2% of the bandwidth overhead of wide-area backhaul networks compared to the standard unquantized FL networks, and 99.6% lower than the centralized cloud pipelines. Empirical Validation Under Non-IID Conditions: Successfully validated on realistic test-sets (5G-NIDD, Edge-IIoTset) under extreme Non-IID data distributions (α = 0.5). The overall threat classification accuracy in the proposed framework was 98.45%, while False Positive Rate (FPR) was low at 1.15%. Real-Time Edge Response and Privacy Guarantees: Achieved a uniformly held formal differential privacy guarantee (ϵ= 2.1, δ= 〖10〗^(-5)) and a sub-3ms edge inference latency of 2.85ms using calibrated L2-norm gradient clipping and Gaussian noise injection. Fully Peer-to-Peer (Decentralized) Federated Topologies: Reducing the need for any "federated coordinator" by adopting the fully decentralized blockchain-based or gossip-protocol aggregation topology, which eliminates risks of having a single point of failure. Asynchronous Aggregation for Dynamic Telecom Nodes: Work on developing asynchronous parameter aggregation algorithms to manage intermittent connectivity and latency variations among very-moveable edge nodes (such as vehicular gateways, or UAV-assisted base stations). Self-Supervised and Zero-Shot Learning Adaptations: Incorporating self-supervised contrastive learning at local edge sites to improve zero-shot recognition of highly subtle, low-volume attack patterns before global aggregation rounds occur. Hardware Acceleration on Neuromorphic Edge Processors: Porting local INT8 neural inference models onto specialized Edge TPU and Neuromorphic hardware to reduce local training CPU utilization below 15%.

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